

Homework 1: What the Judges Actually Score

Paper track. Original scenario, invitational format. Do this on your own.

The Scene

At a weekend robotics tournament, the club's cash bag (about \$600 in registration fees) disappears from the check-in table between 8:15 and 8:40 a.m. The table stands on a strip of exposed dirt just inside the gym's side door.

Investigators recover: a smear of **white powder** on the table edge that **turns iodine blue-black** and does **not** fizz with acid; two **short white animal hairs** on the cash bag strap; a **synthetic (polyester) fiber**, navy blue, caught on the zipper; a small crumb of **dark, fine, iron-rich reddish soil**; and a **partial fingerprint (tented arch)** on the metal clasp.

The Suspects

- **Reese** — runs the snack table (which has flour, powdered sugar, and corn starch out for a baking demo). Wears a navy cotton club shirt. Has a white cat. Was refilling the coffee urn at 8:20 (two witnesses). Fingerprints: ulnar loop, ulnar loop.
- **Sky** — the tournament's setup volunteer; had a key to the cash bag lock. Wears a navy **polyester** windbreaker. No pets. "Checking the field alone" from 8:10 to 8:45. Fingerprints: tented arch, radial loop.
- **Devon** — maintains the school garden; keeps a bin of **reddish iron-rich potting soil** by the gym's side door. Wears a gray fleece. Has a white dog. Fingerprints: plain whorl, ulnar loop.
- **Bailey** — was denied a refund last week and complained loudly to the club treasurer. Runs the 3-D-printing booth (lots of PLA plastic filament, no powders). Wears a green wool sweater. No pets. Fingerprints: plain arch, tented arch.

Problems

1. (5 pts) Complete the evidence grid. For each item: which suspect(s) does it point to, and is it a **strong** (unusual/specific) or **common** match?

Evidence	Points to	Strong or common?
White powder — iodine-positive, no fizz		
Short white animal hair		
Navy polyester fiber		
Reddish iron-rich soil		
Tented arch fingerprint		

- 2. (2 pts)** Which suspect has the strongest **motive**? Do the physical evidence and that suspect line up? What does that tell you?
- 3. (10 pts)** Write a full analysis: name the culprit; give a motive; list the evidence against them, each with a reason; give a timeline note; and clear each of the other three suspects in a sentence.
- 4. (1 pt)** The white powder is iodine-positive and doesn't fizz. Name a powder it could be, with its formula.
- 5. (2 pts)** Explain why "the tented arch fingerprint matches Sky and Bailey" is *class* evidence, not proof.

Answer Key

1.

Evidence	Points to	Strong/common
White powder, iodine +, no fizz	Reese (has corn starch / flour at the snack table)	strong — Bailey has no powders; a starch powder is specific
White animal hair	Reese (cat) or Devon (dog) — need species ID to split	common
Navy polyester fiber	Sky (polyester windbreaker) — Reese's navy shirt is <i>cotton</i> , not polyester	strong — the fiber class rules Reese out for this item
Reddish iron-rich soil	Devon (reddish iron-rich potting soil bin by the side door)	strong — distinctive color/composition
Tented arch print	Sky or Bailey (both have a tented arch); excludes Reese and Devon	class evidence

2. Bailey has the strongest motive (denied a refund, complained). But the physical evidence doesn't point to Bailey — no powder (Bailey has only plastic filament), a wool sweater (not the polyester fiber), no pet (the animal hair). The best motive belongs to a suspect the evidence clears. Say that explicitly.

3. Model answer:

- **Culprit: Sky.** Motive: had a key and the only unsupervised window (8:10–8:45, "checking the field alone" — no witness).
- Evidence: navy **polyester** fiber on the zipper matches Sky's polyester windbreaker (Reese's navy shirt is cotton); **tented arch** print on the clasp is consistent with Sky and excludes Reese and Devon; had the **key** and **no alibi**.
- Timeline: Sky alone 8:10–8:45, covering the whole theft window; Reese was refilling coffee at 8:20 with two witnesses.
- Clear the others:
 - **Reese** — the iodine-positive powder and a white pet match, but the fiber is polyester (Reese wears cotton), and Reese has a witnessed alibi at 8:20.
 - **Devon** — the reddish soil matches Devon's potting-soil bin, and Devon has a white pet, but there's no powder link, no fiber match, and the tented arch print excludes Devon (whorl/loop). The soil could be secondary transfer — the bin is right by the door everyone used.
 - **Bailey** — strongest motive but **no** matching physical evidence (plastic filament not powder, wool not polyester, no pet). Motive without evidence is not proof.

4. Corn starch — $(C_6H_{10}O_5)_n$. (Flour also acceptable.)

5. Lots of people have a tented arch (~a few percent of prints, but that's still millions of people), and a partial print can't be matched on minutiae. It narrows the field (rules out anyone without a tented arch) but doesn't identify one person the way a full minutiae match or DNA would.